Classification and Evaluation Metrics

LumoraAI Presentation Demo • 04 Machine Learning

Classification

• Classification predicts a discrete class label. Binary classification has two classes, while multiclass classification has more than two mutually exclusive classes in the simplest formulation.

Confusion matrix

• For binary classification, true positives and true negatives are correct predictions; false positives and false negatives are errors. Precision measures the fraction of predicted positives that are actually positive, while recall measures the fraction of actual positives that were retrieved.

Thresholds

• For probabilistic classifiers, changing the decision threshold changes the balance between false positives and false negatives. The appropriate threshold depends on the cost of each type of error.

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